# TCG-Net — Results Summary

Two-branch cooperative game-theoretic fusion of resting-state functional connectivity and plasma proteomics · UK Biobank

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All results below are held-out test-set numbers. Joint-model figures are aggregated over **11 independent training runs** differing only in random seed, each trained to convergence under its own early-stopping criterion. Participants used to evaluate the joint model are excluded from every pretraining and baseline fit, so no evaluation subject influences any model it is scored on.

## 1. Fluid intelligence — current results

The joint model is trained on 3,471 participants and evaluated on 747 held-out participants. Alongside the fused prediction it exposes three internal predictions: an fMRI-only head, a proteomics-only head, and a plain MLP on the concatenated embeddings. Scoring all four on the same subjects shows where the performance originates.

| Output | $R^2$ | $R^2$ calibrated | Pearson $r$ |
|---|---|---|---|
| **Fused prediction (full model)** | **0.0725 ± 0.0056** | **0.0768** | **0.279** |
| fMRI branch only | 0.0652 ± 0.0028 | 0.0665 | 0.260 |
| Proteomics branch only | −0.0010 ± 0.0022 | 0.0285 | 0.172 |
| Concatenation head (naive fusion) | 0.0665 ± 0.0060 | 0.0706 | 0.267 |

**The calibrated column matters.** An affine rescaling fitted on the validation fold and applied to the test fold corrects a systematic amplitude error in the model outputs. It changes the reading of the proteomics branch entirely: a raw $R^2$ of approximately zero becomes 0.0285, essentially the theoretical maximum for a predictor correlating at $r$ = 0.17. The proteomics branch is therefore **extracting all of the signal available to it** — its raw score of zero reflected a collapsed output scale, not an absence of information.

### Comparisons between the model's own outputs

| Contrast | Mean Δ $R^2$ | Seeds in favour | $t$ (df = 10) |
|---|---|---|---|
| Fused − fMRI branch | +0.0073 | 9 / 11 | +3.82 |
| **Fused − fMRI branch (calibrated)** | **+0.0103** | **11 / 11** | — |
| **Concatenation head − fused** | **−0.0060** | **1 / 11** | **−4.19** |

The cooperative fusion output exceeds its own fMRI branch on 11 of 11 seeds once calibrated, and exceeds the naive concatenation head on 10 of 11. Within the architecture, the Shapley-guided gate outperforms both the single-modality path and the standard concatenation alternative.

### Why the proteomic contribution is small

The proteomics-to-cognition association decomposes cleanly, and the loss is attributable rather than anomalous.

| Setting | Pearson $r$ | Explanation |
|---|---|---|
| Full proteomics cohort, baseline visit | 0.345 | reference |
| Imaging subcohort, baseline visit | 0.254 | cohort restriction |
| Imaging subcohort, imaging visit | 0.171 | ~ 9-year blood-to-scan interval |

The final step is fully accounted for by measurement attenuation: the test–retest correlation of the cognitive measure between the two visits is 0.636 on these participants, and 0.254 × 0.636 = 0.162 against 0.171 observed. A predictor correlating at $r$ = 0.17 is capped at $R^2 = 0.029$ however it is modelled. **This is a property of the cohort design** — plasma was sampled roughly nine years before the scan — **and not of the model.**

## 2. Ablation and baseline testing — fluid intelligence

Every model below is evaluated on the same 747 held-out participants. Two families of linear baseline were run. The first is fitted on the joint training fold only (3,471 participants). The second is given the same advantage the deep encoders receive: fitted on the full single-modality cohort with all joint validation and test participants excluded, then adapted to the joint training fold by a two-parameter affine map. The second family is the fair comparison.

| Model | Fitted on | $R^2$ | Pearson $r$ |
|---|---|---|---|
| Predict the training mean | — | −0.0007 | — |
| Ridge, proteomics (2,920 features) | 3,471 | 0.0242 | 0.165 |
| Ridge, connectivity (2,145 edges) | 3,471 | 0.0409 | 0.213 |
| Ridge, concatenated features | 3,471 | 0.0317 | 0.183 |
| Ridge, proteomics — pretrained | 11,528 | 0.0363 | 0.195 |
| Ridge, connectivity — pretrained | 30,511 | 0.0818 | 0.290 |
| **Late fusion of the two pretrained ridges** | both | **0.1035** | **0.325** |
| TCG-Net, fused output | 3,471 | 0.0725 | 0.279 |
| **TCG-Net, fused output (calibrated)** | 3,471 | **0.0768** | — |

### What the baselines establish

**(a) Multimodal integration genuinely helps.** Late fusion of the two pretrained linear models reaches 0.1035 against 0.0818 for connectivity alone — a gain of +0.0217. Plasma proteomics does add information beyond the connectome for cognitive prediction.

**(b) Feature-level concatenation is not the way to combine them.** Fitted on the joint training fold, concatenating raw features (0.0317) performs **worse** than connectivity alone (0.0409). Naive early fusion destroys signal at this sample size, which motivates a learned fusion mechanism.

**(c) The current network does not yet beat the strongest linear baseline.** Late fusion of pretrained ridges reaches 0.1035 against 0.0768 for the calibrated fused output — a shortfall of 0.027 $R^2$, several times the seed-to-seed standard deviation of 0.006.

> The likely explanation is **representational rather than architectural**. The linear model uses all 2,145 connectivity edges with weights estimated from 30,511 participants, whereas the graph encoder compresses the connectome to 128 dimensions and its downstream heads are refitted on 3,471. The fusion mechanism itself performs well in every internal comparison (Section 1); it is the encoded representations that trail the raw features.

**Ablations still outstanding.** Single-modality, early-fusion and late-fusion comparisons are covered above. Not yet run: attention-based fusion, a learned non-Shapley modality gate, and the model without the interaction term, the squeeze-and-excitation block, or the graph encoder. The learned non-Shapley gate is the most informative of these, since it isolates the contribution of the game-theoretic weighting specifically.

## 3. Ablation and baseline testing — age

Age was assessed as a second target because both modalities carry substantial signal for it, which fluid intelligence does not permit.

### Single-modality proteomics encoder

| Model | $R^2$ | MAE (years) | Pearson $r$ |
|---|---|---|---|
| Predict the training mean | −0.0000 | 6.99 | — |
| **Ridge, all 2,920 proteins** | **0.8690** | **2.35** | **0.932** |
| Ridge, 200 strongest proteins | 0.7677 | 3.14 | 0.876 |
| **Proteomics encoder (Stage 1)** | **0.868** | — | — |

Evaluated on 7,904 held-out participants, **the proteomics encoder and ridge regression are indistinguishable** (0.868 against 0.8690). Age is linearly decodable from an Olink panel, so this figure reflects the information content of the assay rather than the encoder architecture. This was the specific comparison requested, and the answer is that the proteomics branch does not require a deep architecture.

### Joint cohort — the multimodal baselines

Evaluated on 799 held-out participants from the joint cohort, using the same two baseline families as in Section 2.

| Model | Fitted on | $R^2$ | Pearson $r$ |
|---|---|---|---|
| Predict the training mean | — | −0.0001 | — |
| Ridge, connectivity | 3,730 | 0.2469 | 0.498 |
| Ridge, proteomics | 3,730 | 0.7742 | 0.880 |
| Ridge, concatenated features | 3,730 | 0.7828 | 0.885 |
| Ridge, connectivity — pretrained | 33,081 | 0.3526 | 0.595 |
| Ridge, proteomics — pretrained | 35,754 | 0.8106 | 0.901 |
| **Late fusion of the two pretrained ridges** | both | **0.8228** | **0.908** |

**The multimodal gain is positive for age.** Late fusion improves on proteomics alone by +0.0122 $R^2$, and unlike fluid intelligence, concatenating raw features also helps rather than hurts (0.7828 against 0.7742). Both modalities carry real age information and are partially complementary — although the connectome's contribution is largely redundant with the proteome's, since only about 3.5% of its standalone signal survives as independent.

Pretraining is worth a great deal on the imaging side specifically: connectivity ridge improves from 0.2469 to 0.3526 when fitted on 33,081 rather than 3,730 participants. This is why the pretrained baseline family was constructed — comparisons against joint-fold-only baselines systematically overstate any deep model's advantage.

### Status of the joint age model

The two-branch model has not yet been run on age. It requires a connectome encoder pretrained against age, which is pending on the imaging side. The tables above therefore establish the threshold rather than a comparison:

> The joint age model must exceed $R^2 = 0.8228$ ($r$ = 0.908) to demonstrate value over a purely linear approach.

## Summary

The pipeline now produces properly held-out, multi-seed, leakage-controlled results on both targets. Three findings stand out.

1. The cooperative fusion mechanism outperforms both single-modality prediction and naive concatenation within the architecture, consistently across seeds.
2. Multimodal integration is genuinely informative for both targets. The proteomic contribution to cognition is small for a quantifiable reason — a nine-year interval between blood sampling and scanning that attenuates the association to a ceiling of $R^2 = 0.029$.
3. Neither deep encoder currently outperforms a regularised linear model given the same training data, which locates the remaining work in the learned representations rather than in the fusion mechanism.

**Immediate priorities.** Complete the remaining ablations, particularly the learned non-Shapley gate; obtain an age-trained connectome encoder so the joint age model can be evaluated against the 0.8228 threshold; and test the branch comparisons at the participant level by bootstrap, since the figures reported here quantify variation across training runs rather than across participants.